

MS II

Moshood Abiola Garuba

2025-03-19

Question 7e:

```
# Given parameters
n <- 50
p <- 1/3 # Common probability for geometric distributions

# Expected values (means) of geometric distributions
mu_X <- 1 / p
mu_Y <- 1 / p
mu_Z <- 1 / p

# Variances of geometric distributions
sigma2_X <- (1 - p) / p^2
sigma2_Y <- (1 - p) / p^2
sigma2_Z <- (1 - p) / p^2

# Compute mean and variance of L
E_L <- mu_X + mu_Y + mu_Z
Var_L <- (sigma2_X + sigma2_Y + sigma2_Z) / n
SD_L <- sqrt(Var_L) # Standard deviation of L

# Compute P(L > 10) using normal approximation
P_L_10 <- pnorm(10, mean = E_L, sd = SD_L)

# Print the probability
print(P_L_10)
```

```
## [1] 0.9522096
```

Question 15:

(i) Probability that the Sample Variance of Company A Exceeds 0.65 Amp

```
pchisq((0.65 * 9 / 0.04), df = 9, lower.tail = FALSE)
```

```
## [1] 5.269854e-27
```

(ii) Probability that the Sample Standard Deviation of B is 2/3 Times Larger than That of A

```
pf(4/9, df1 = 14, df2 = 9, lower.tail = FALSE)
```

```
## [1] 0.9159394
```

(iv) 75th Percentile of the Distribution of S^{2a}/S^{2b}

```
qf(0.75, df1 = 9, df2 = 14)
```

```
## [1] 1.469714
```

Question 17

```
# Load necessary library
library(stats)

# Given parameters
n <- 5 # Sample size
m <- 3 # Binomial parameter (number of trials)
p <- 0.34 # Probability of success

# Compute the probability mass function (PMF) of  $X \sim \text{Binomial}(3, 0.34)$ 
pmf_values <- dbinom(0:m, size = m, prob = p)

# Compute  $P(X_{(3)} \geq 3)$ 
#  $X_{(3)}$  is the third largest value in the sample
# We need  $P(X_{(3)} \geq 3) = 1 - P(X_{(3)} \leq 2)$ 

# Compute probability that at most 2 values in the sample are below 3
threshold_prob <- sum(pmf_values[1:3]) #  $P(X \leq 2)$  from the Binomial PMF
prob_X3_leq_2 <- sum(dbinom(0:2, size = n, prob = threshold_prob))

# Compute  $P(X_{(3)} \geq 3)$ 
prob_X3_geq_3 <- 1 - prob_X3_leq_2

# Display the result
cat("P( $X_{(3)} \geq 3$ ) =", round(prob_X3_geq_3, 4), "\n")
```

```
## P( $X_{(3)} \geq 3$ ) = 0.9994
```